

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

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Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

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Introduction

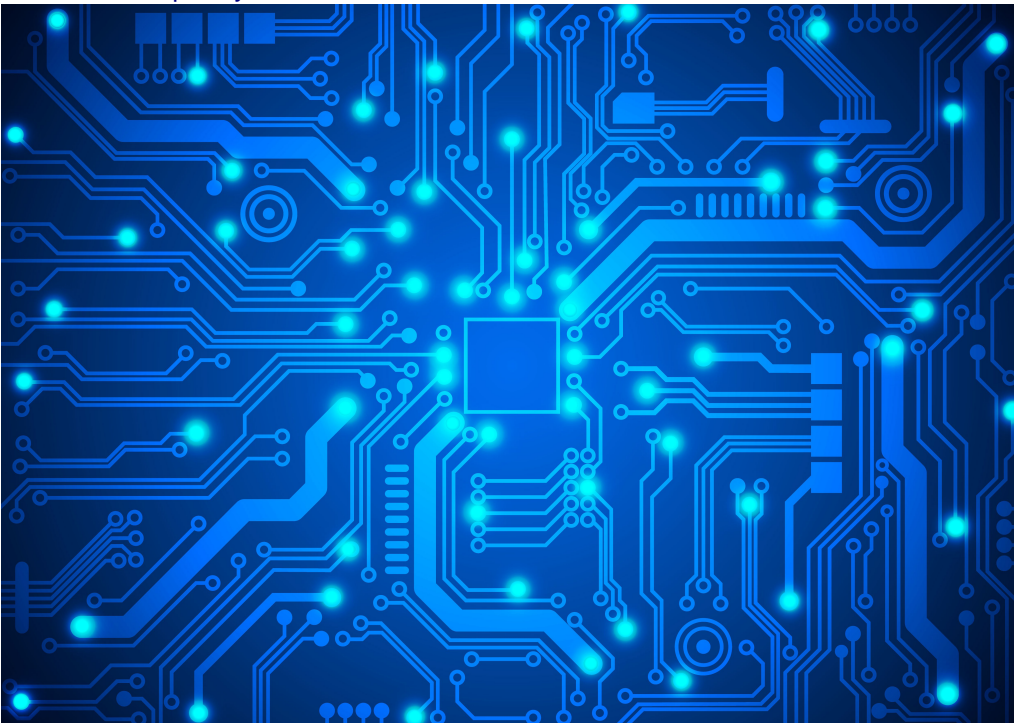


Introduction



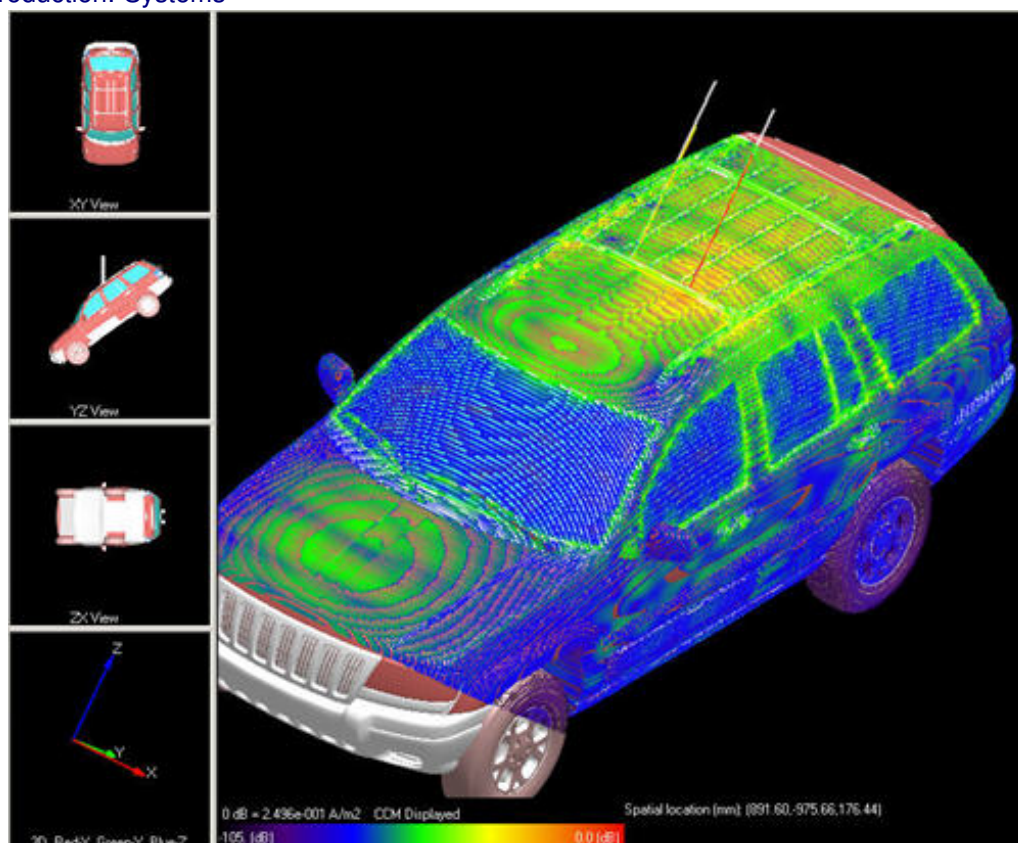
1.4

Introduction: Complexity



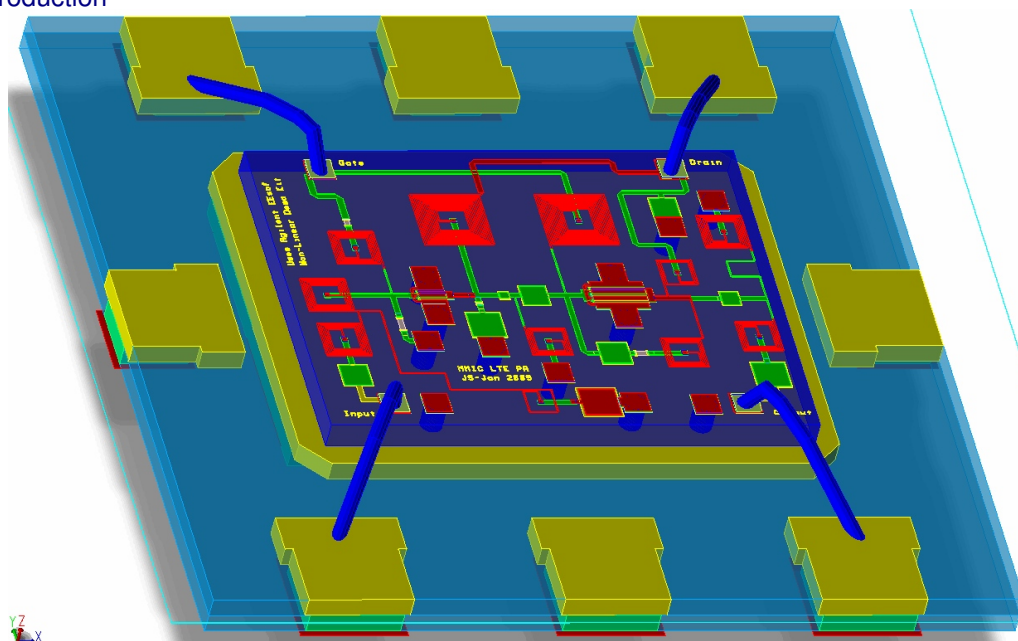
1.5

Introduction: Systems



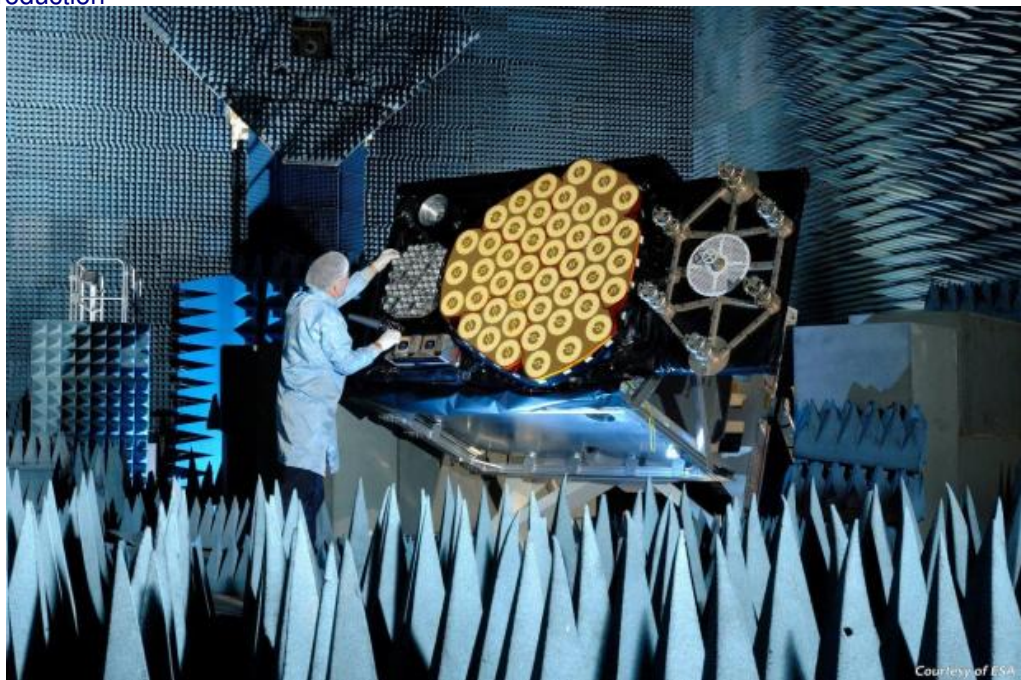
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

1.9

Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

1.10

Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

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Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

1.12

1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

1.13

Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

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WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

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WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

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Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

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Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

1.19

Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: bash and tcsh

Start: ads after using ssh to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

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Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

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Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

1.22

The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

Part I

Part 2: ADS Circuit Design

2 Small-Signal Amplifiers

Feedback 2 from correction

- Line: resonances due to:
- Line of a certain length
- : Line at $\lambda/4$ with open: acts like a short at the input, for $n= 1,3, 5$,
- : Line at $\lambda/2$ with open: acts like on open at the input...
- With the change of the frequency you go through the wavelengths (in the material)
- Scaling to -300 dB ? 30 order of magnitude !!!!

6.25

Orders of magnitude in unit of human being

- Proton 10^{-15}
- Atom 10^{-10}
- You 1 (2 m)
- Small town 10^3
- Earth radius (6370 km) 10^7
- Earth - Moon (383.000 km) 10^8
- Radius sun (700.000 km) 10^9
- Radius of earth orbit (150.000.000 km) 10^{11}
- Sun system (6.000.000.000km) 10^{13}
- Next Fix star 10^{17}
- Center of the milky way 10^{20}
- Next galaxy 10^{23}
- Radius of the visible universe 10^{26}

Source: www.leifiphysik.de

6.26

Today: What is design: The Design Flow

- Design: multiple targets
- What to do first
- and in which order...

6.27

Educational targets

- Perform a conjugate complex matching at input and output through the selection of matching networks.
- Design an input and output network with lumped elements.
- Conjugate-complex matching as one form of useful small-signal matching.
- However, directly understand, that maximizing gain is not the only choice which can be useful from the small-signal side.
- Understand noise matching in a trade-off to conjugate complex matching.
- Understand, that, no matter what matching I choose, both matchings have to obey the superior boundaries of stability. You can only address the optima of gain or noise or both, when these impedances you choose fulfill stability criteria. In some cases, matching both input and output conjugate-complex will lead to an instable amplifier, so you need to do something about it.

6.28

Educational targets II

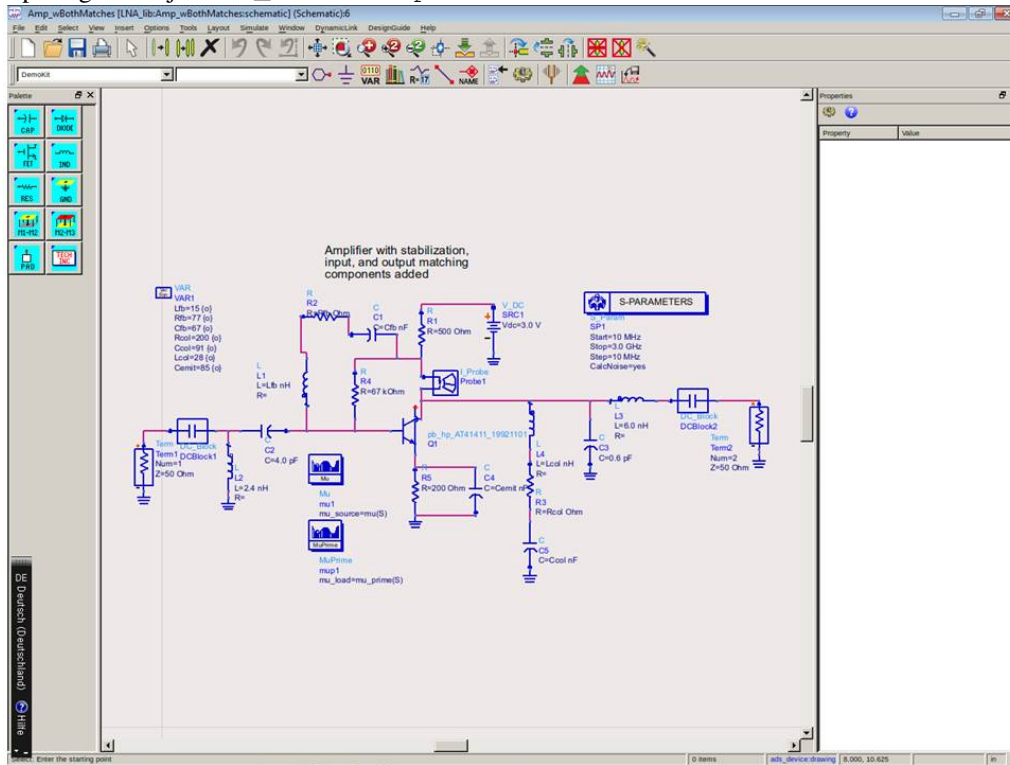
- Understand for the first time, that stability is an issue to be discussed in parallel at input and output of the device, i.e., it is an iterative designing: design flow.
- RELAX: stability is one of the most complex issues of RF-, you have to get a first feeling, and you will gain experience in it.

6.29

2.1 Small-Signal Matching

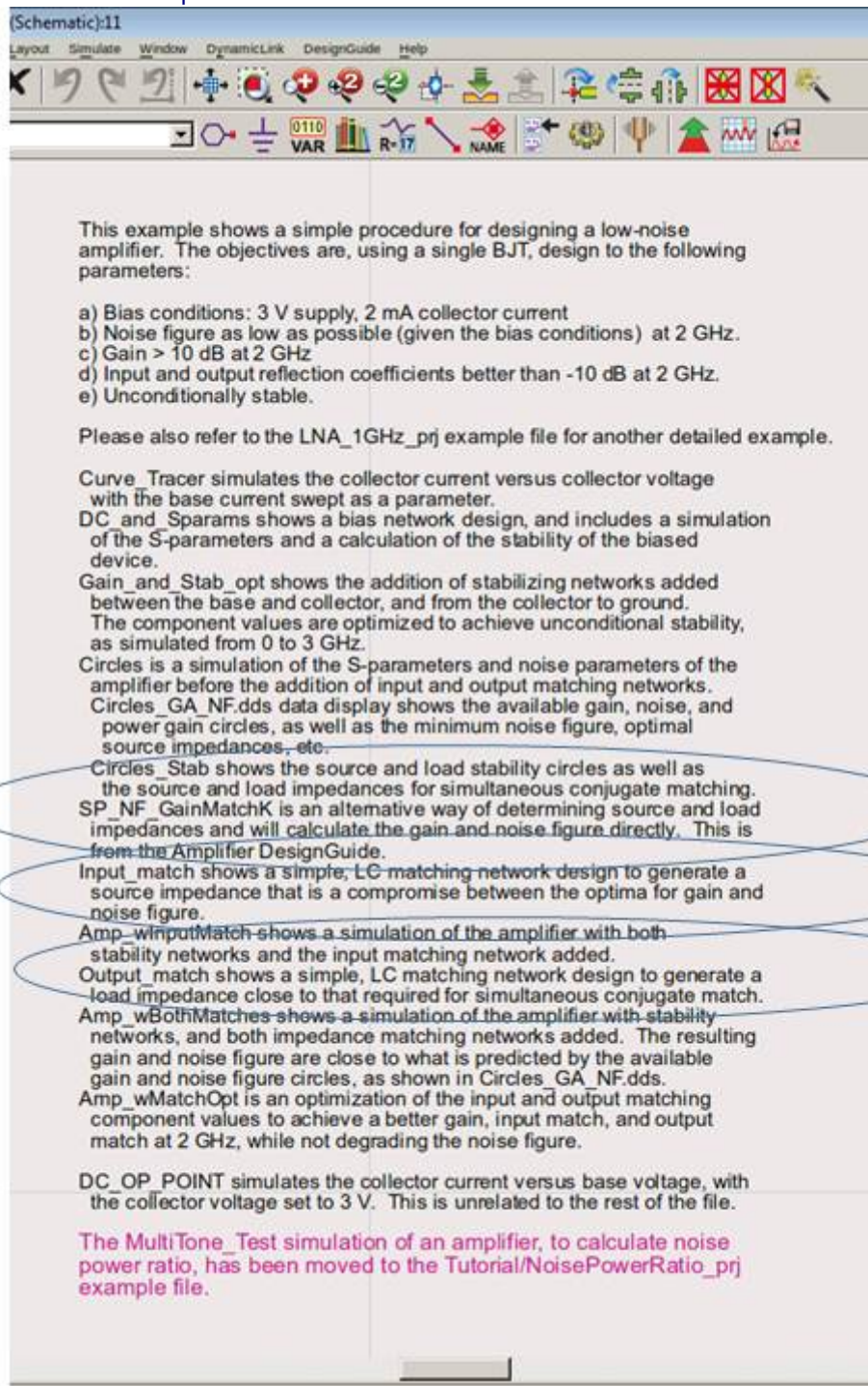
Small-Signal Matching

Open again Project: MW_CktsLNAzap



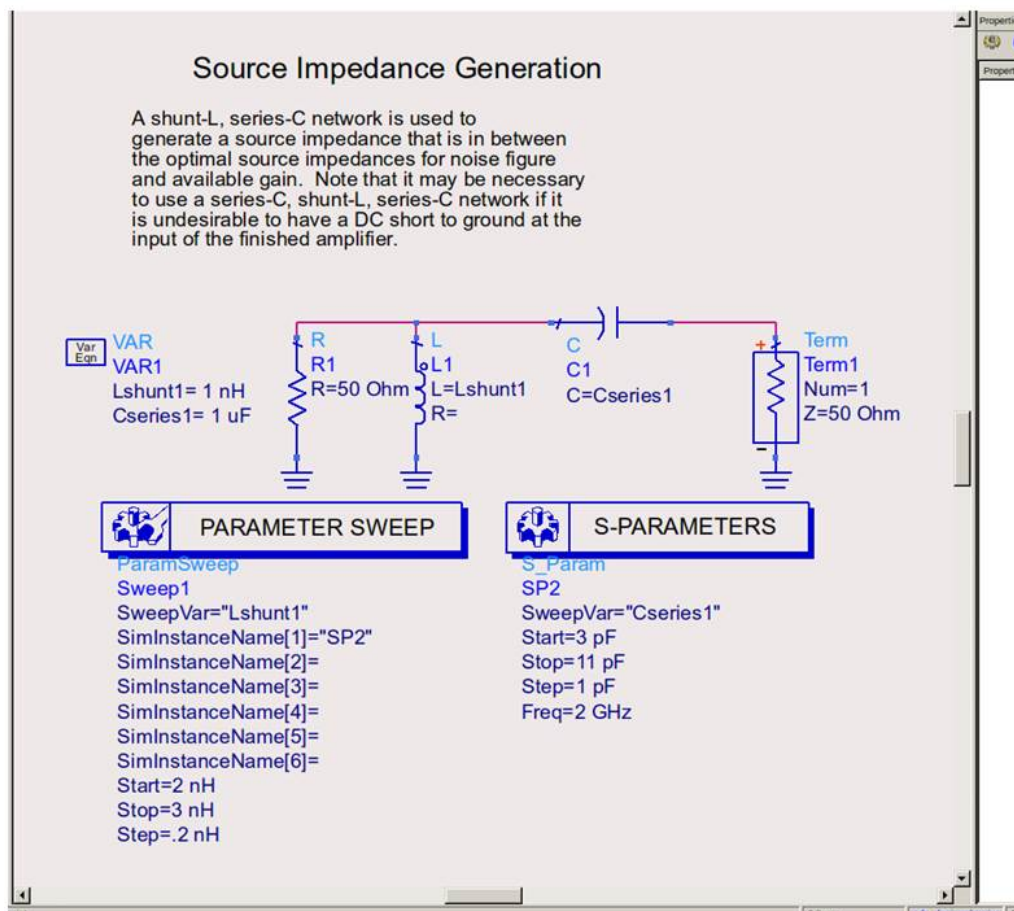
6.30

Readme of the example



2.2 Input Matching

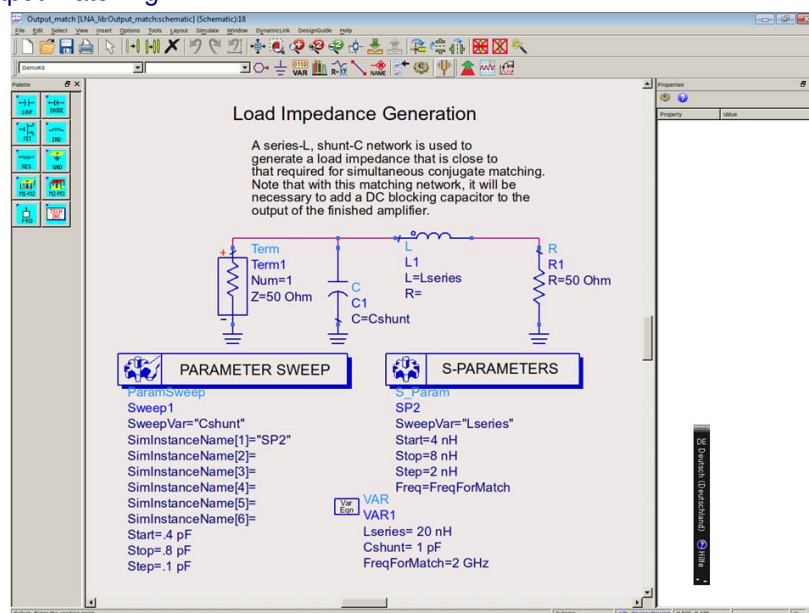
Match for maximum gain: conjugate-complex matching



6.32

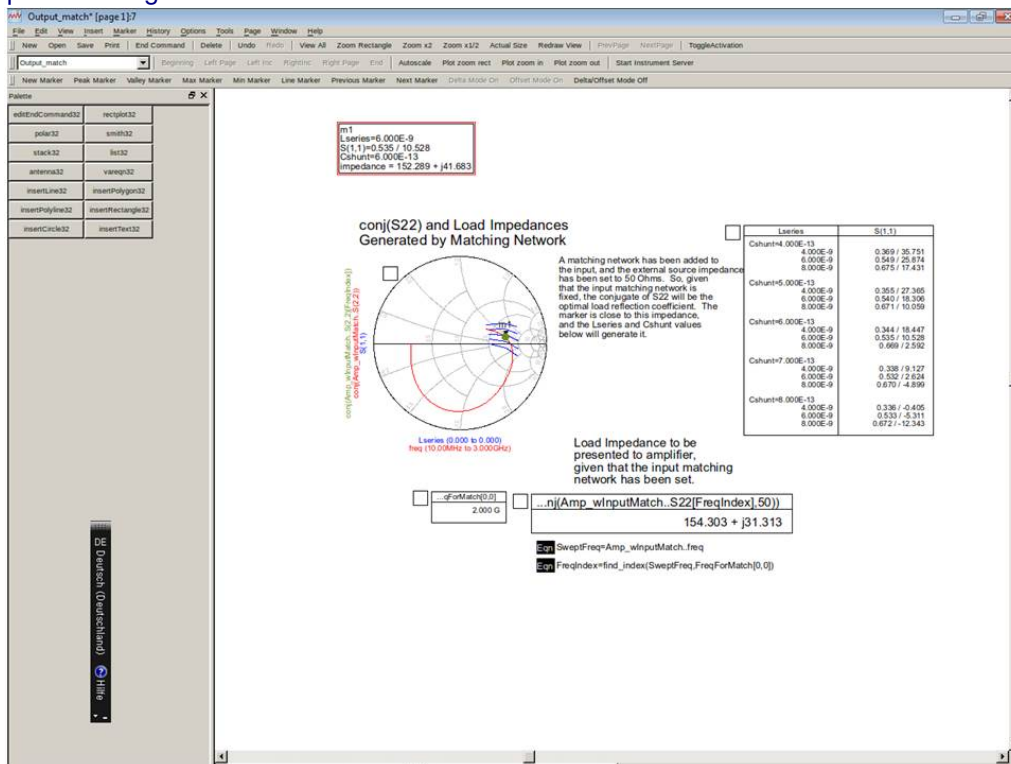
2.3 Output Matching

Output Matching



6.33

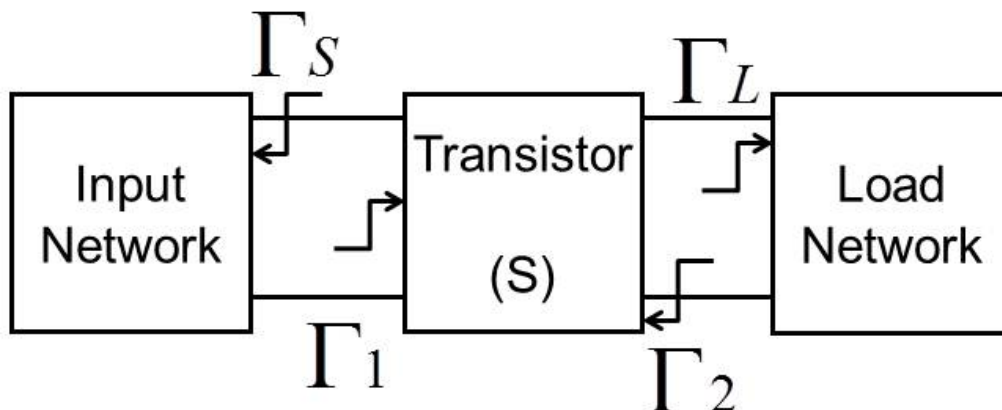
Output Matching II: Results



2.4 Stability Analysis

Stability Analysis: Principles

What happens if I arbitrarily tune input and output for a matching of my choice, e.g., maximizing gain.

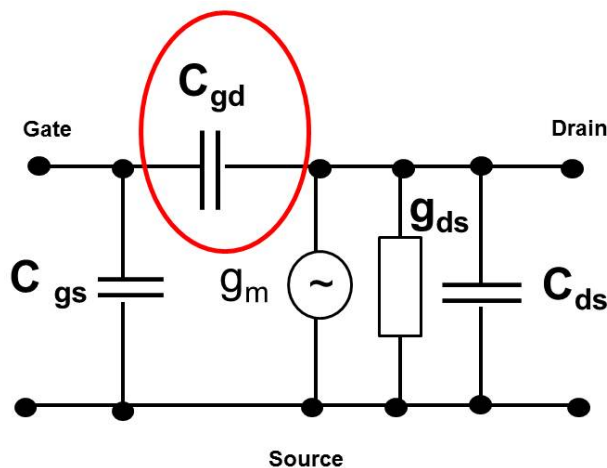


Answer: I and you will fail hopelessly !!!!

6.35

Oscillators from Amplifiers

Reason for stability problems: Built-in Feedback in any active Device, i.e., any active device feeds a fraction of the power from the output to the input via C_{gd} as there is no ideal isolation between output and input.

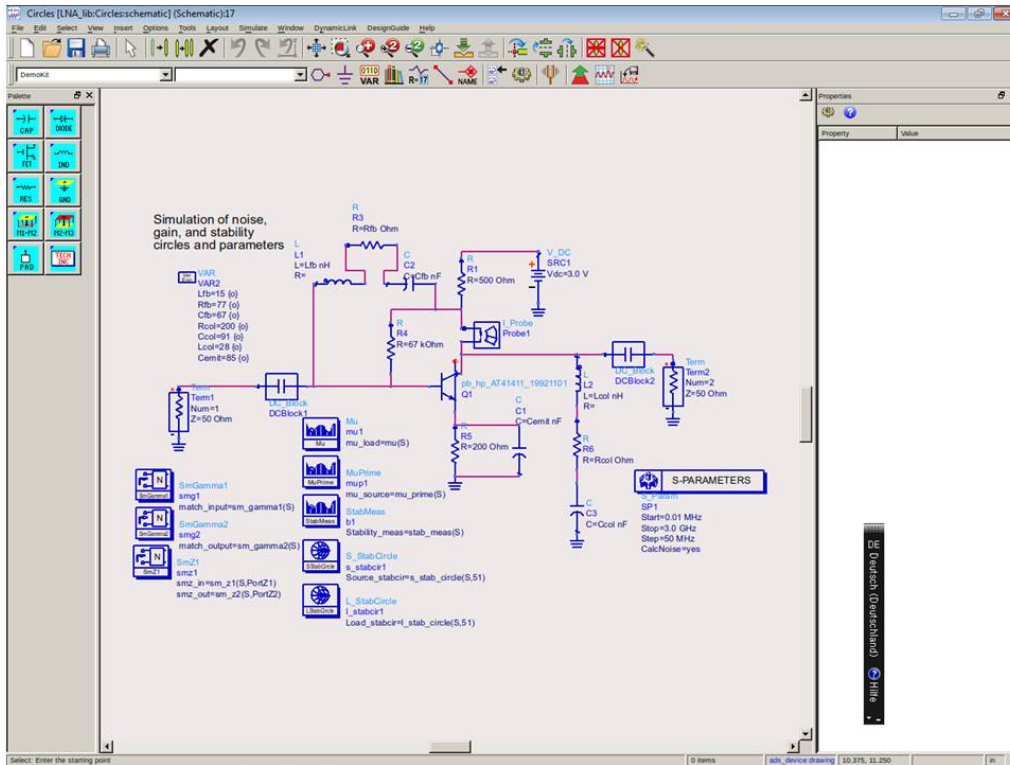


FET: Intrinsic feedback capacitance C_{gd} , Bipolar: C_{bc} or C_{jc} : both devices are not-unilateral

6.36

Stability Analysis

The example gives a way to plot stability circles and stability factors called μ . The stability circles are regions in the Smith Chart, which tell you something about instable and stable impedances.



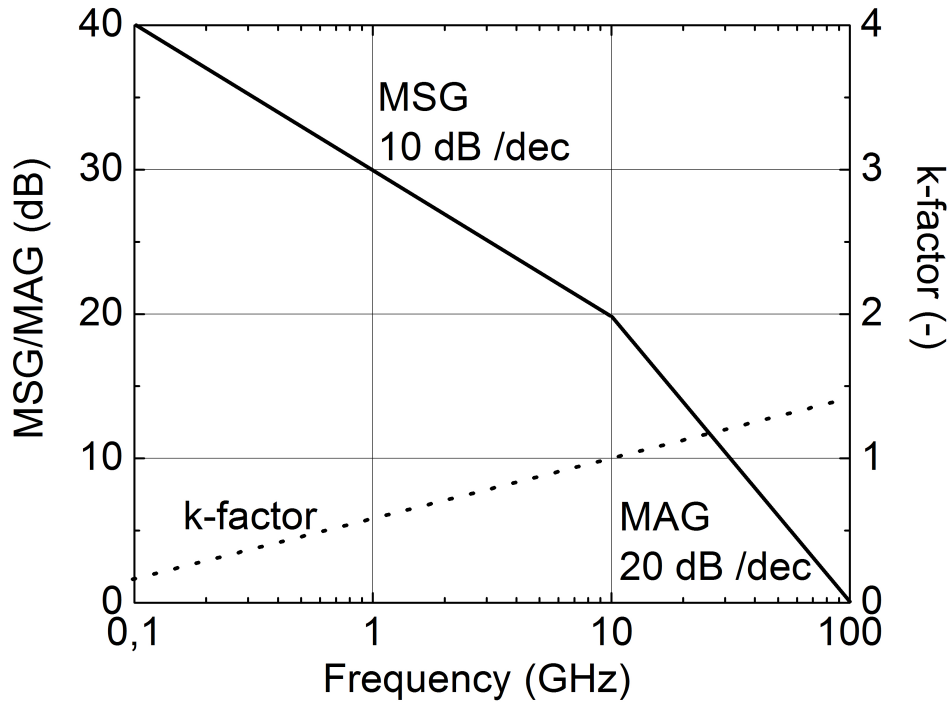
6.37

Stability Analysis II

This possibility of the intrinsic destabilization is only happening up to a certain frequency. It is related to the gain of an active device. In the graph below this is where the stability factor of the device k hits 1, i.e., here 10 GHz, further there is a gain MSG (maximum stable gain), which tells us, how much gain can I get once the device is stable (this makes sense, as an INSTABLE device has a power gain of infinity as you are getting an output signal without providing an input power). Beyond 10 GHz (in this case), you cannot any more destabilize the device by tuning input and output for principal reasons and you are getting the maximum gain of MAG (Maximum available gain). So the measures you have to take for stability are only needed to 10 GHz in this plot. In the exercise example it will be a different frequency. Be aware, if you don't know yet, what k , MSG, and MAG are: they are simple functions of the S-parameters measured. They are known once you know the S-parameters and they provide the information on gain and stability. As these are device properties, MSG/MAG are called invariants.

6.38

Stability Analysis II



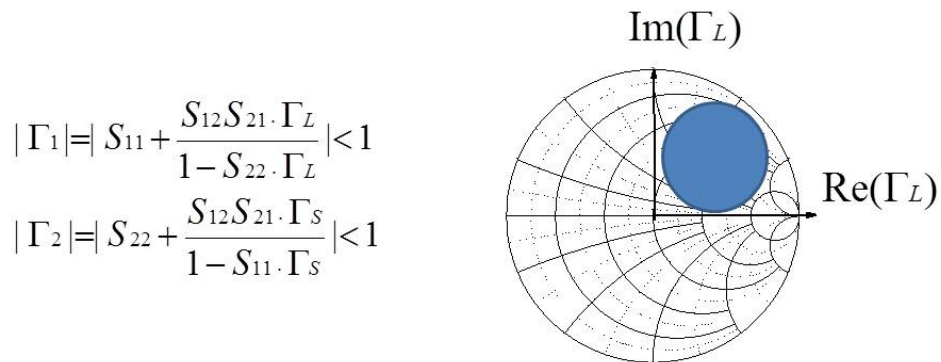
k and MAG and MSG are simple functions of the S-parameters:

$$MSG = \frac{|S_{21}|}{|S_{12}|} \quad (1)$$

6.39

Stability Analysis IV

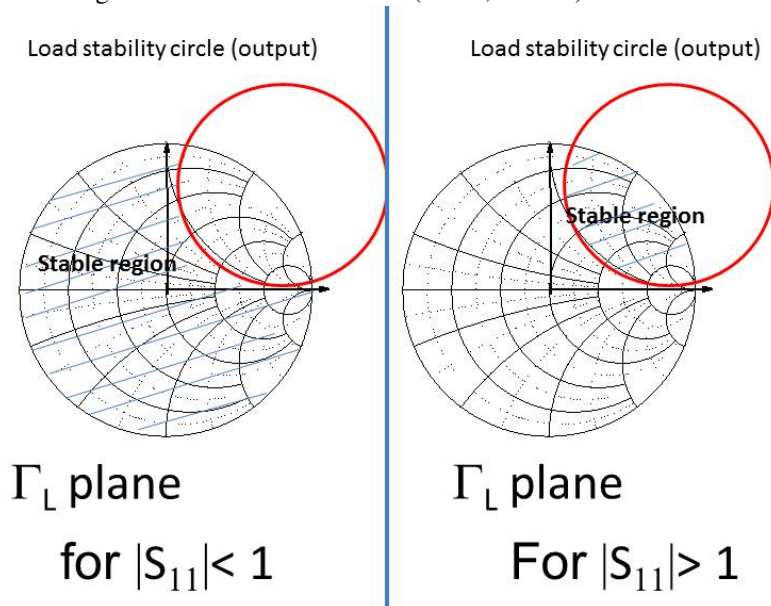
Nature makes it nice for us: Around an input with an S-parameter S_{11} , certain circles can be found. They describe regions inside and outside of these circles, which are helpful for the analysis of stability



6.40

Stability Analysis V

Which impedance region is now stable or instable (inside, outside) ?

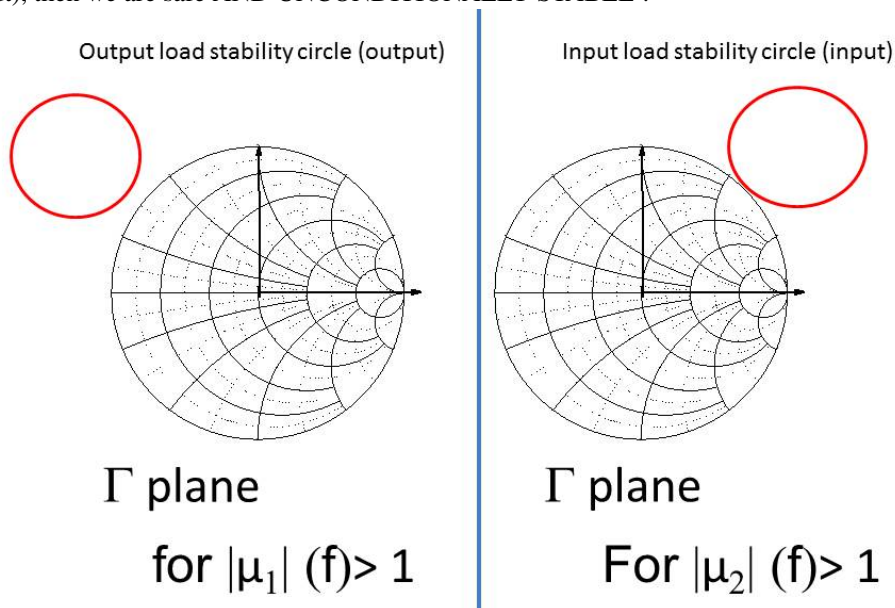


Understand: Whether an impedance is stable at the output, depends on what you do at the input !!! **If you say this is too complex for me; yes it is, but it assures that designers keep their jobs !!!**

6.41

Stability Analysis vs. Frequency: μ

μ is another useful function which defines the closest point (radius) of instability from the center of the Smith Chart and can be easily viewed vs. frequency. If $\mu_1(f)$ (output) and $\mu_2(f) > 1$ (input), then we are safe AND UNCONDITIONALLY STABLE !



6.42

2.5 Input Low-noise Matching

Noise-Matching

1. Conjugate-complex matching optimizes for gain in a small-signal matching. It is the golden rule of matching for GAIN.
2. Noise matching is another, additional and independent requirement for small-signal matching for some (not all) applications. It serves here to show you, that conjugate-complex matching is not the only possible solution to match active devices, but that additional requirements will enter your thoughts pretty soon. So there will be always **trade-offs**.
3. In this case of additional noise optimization some gain is sacrificed to optimize i.e., minimize a second independent quantity, i.e., RF-noise for the same frequency.
4. This requires slightly different impedances than optimum gain, i.e., for noise we are deviating slightly from the conjugate-complex matching
5. In both cases, searching for maximum gain or minimum noise or both, I have to check for stability criteria in order to make sure that I am building an amplifier and NOT an oscillator !

6.43

3 Tasks 10: Small-Signal Circuit Design (Block 3 task 1)

3.1 Tasks 10: Small-Signal Circuit Design (Block 3 task 1)

Tasks 10: Small-Signal Circuit Design (Block 3 task 1), 40 points

1. Explanation: Go to the example: LNA in examplesMWCKTS
2. Explanation: Open the readme.
3. Explanation: Concentrate first on the three schematics named input match, output match, and circles stab in amp design. Be careful in the following: the device in this example is not designed for conjugate-complex matching alone. The nature of this task is understand conjugate complex matching in relation to what needs to be done **beyond** simple conjugate complex matching at input and output to achieve REAL circuits.
4. Task: Goto DC_andS-params:: Plot the magnitude of the S-parameters S12 (reverse isolation) going from -35 dB to 0 dB. Explain the behavior of S12 over frequency (2 points for the plot)?
5. Task: How can you influence the feedback in the present design: Hint: what is also contributing to the feedback (Four sentences max.) (3 points) ?

6.44

Tasks 10: Small-Signal Circuit Design (Block 3 task 1)

6. Task: Look in to the schematic Amp_wBoth : Task: In the schematic: check at least two different networks (plot them) and tell me their main function in terms of biasing (base, collector), feedback, matching, decoupling...In case there are double functions of the above components, do not get nervous, mention it (2 plots and one table: network and function)(5 points for the two answers)
7. Explanation; Look at the DC and Sparams schematic. Task: Find and plot the real part of the impedance (in Ohms) for conjugate-complex matching of the input of the device for 100 MHz... 6 GHz (just the Plot).
8. Plot the S-parameters (magnitude/phase) at the input of the device (S11) and in the same Smith diagram of a (theoretical, not to be realized) lumped network, which does conjugate complex match over the same frequency in the **same** Smith chart. This is a very simple task, so dont overdo it. Explanation: It is intended to VISUALIZE the conjugate complex matching (Just the Plot)(3 points)
9. Task: Why can such a network not be realized as you propose it for this frequency range (three sentences max.)(2 points) ?
10. Task: Goto circles Gain and stab opt: a.) Plot the stability circles (output and input) at 1.2 GHz and at 2.6 GHz in circles for VCE = 1.2 V. Use the slider (2 plots, 5 points for each plot)

6.45

11. Task: Explain which is now the stability situation (stable /unstable) for the conjugate complex circuit at 1.3 GHz. Find one setting for the matching, where the device becomes unstable at 1.3 GHz (just the plot, plus 2 sentences, 5 points) Explanation: an easy way in reality to do that is to play with the biasing networks which are already included.
12. Task: Explain the usefulness of mu load and mu source. Check help for the explanation, task: simulate up to 3.0 GHz (Plot plus max. 3 sentences) (5 points)
13. Task: Explain the stability measure shortly. Just for you in one sentence: Why are these measures (Mu and stability measure) easier to catch than the graphical idea of the stability circles ? (Give three sentences on the advantages)(5 points)

6.46

Tasks 10 format for 40 points

1. 10.4 Give plot with the right scaling... (2 point)
2. 10.5 Give four sentences max...(3 points)
3. 10.6 2 plot and one table (5 points)
4. 10.7 Just the plot (5 points)
5. 10.8 Give just the plot ! (3 points)
6. 10.9 Give three sentences max. (2 points).
7. 10.10 Give just 2 plots (5 points)
8. 10.11 Give one plot plus two sentences (5 points)
9. 10.12 Give one plot and three sentences (5 points)
10. 10.13 Give three sentences max. (5 Points)

6.47